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Coupling based estimation approaches for the average reward performance potential in Markov chains☆

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ABSTRACT

Performance potential is an important concept in the sensitivity analysis of Markov chains. The estimation of performance potential provides the basis for the simulation-based optimization and sensitivity analysis of Markov chains. In this study, we present novel estimation approaches for the average reward (or cost) performance potential by combining perturbation realization factors and coupling techniques for Markov chains with finite state space. These approaches can effectively implement estimation with geometric variance reduction for average reward performance potential. Meanwhile, a number of coupling methods, including two optimal coupling methods, can be applied to further reduce estimation variance or simulation time. The numerical tests show that our approaches can significantly enhance the simulation efficiency.

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1. Introduction

Performance potential (Cao, 2007) is referred to by different terms in various fields. For the average reward (cost) case, performance potential is also called "bias" (Puterman, 1994), "relative cost function" (Bertsekas, 1995), or "value function" (Munos, 2006). Performance potential plays an important role in solving the Markov decision process (Cao, 2007; Jia, 2011; Li & Wu, 2016; Xia & Jia, 2015). The estimation of performance potential provides the basis for the simulation-based optimization (Chang, Fu, Hu, & Marcus, 2007; Cooper, Henderson, & Lewis, 2003; Marbach & Tsitsiklis, 2001) and sensitivity analysis of Markov chains (Cao, 2007). Monte Carlo simulation is a powerful tool for estimating performance potential (Cooper et al., 2003). Regular Monte Carlo estimation can provide variance reduction with order $O(1/N)$, where N is the number of sample paths of a Markov chain. In general, variance reduction simulation techniques include importance sampling, correlated sampling, control variate, stratified sampling and so on Hammerley and Handscomb (1964). Estimations with less variance can be obtained by combining the aforementioned techniques. Estimations with geometric variance reduction (EGVR), the variances of which have a geometric reduction rate, have been investigated by processing the aforementioned methods iteratively (Halton, 1994; Kollman, Baggerly, Cox, & Picard, 1999) and have been applied to continuous-time Markov processes (Gobet & Maire, 2006), discrete-time Markov processes, and gradient estimation (Munos, 2006). As described in Munos (2006), the EGVR of average reward performance potential is subtle and deserves intensive treatment. An estimation approach with geometric variance reduction was proposed in Munos (2006) for average reward performance potential. However, additional variables, such as average reward (also called average expected gain in Munos (2006)) and steady state probability distribution, should be estimated and the sample path functional should be truncated artificially. These problems eliminate the benefit of EGVR and make reducing variance with a geometric rate difficult. A novel generalized fundamental matrix was proposed in Xia and Glynn (2016) to compute average reward performance potential. This fundamental matrix may provide new computation approaches for average reward performance potential; however, it seems to be difficult to find the corresponding sample path functional for EGVR.

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The perturbation realization factor (PRF) proposed in Cao (2007) and Cao and Chen (1997) provides a relative value of performance potentials and can be used to estimate performance potential based on a single sample path (Cao, 2007) and to optimize the performance of queueing systems (Xia & Cao, 2012). In this study, we apply PRF to implement EGVR for average reward performance potential. Coupling methods have been widely used in statistics (Griffeath, 1975; Lindvall, 1992) and simulations. We do not intend to provide a comprehensive survey of coupling methods, and instead, we limit our review to studies that are most relevant to our problem. As a coupling method, common random number (CRN) (Glynn, 1985; Pflug, 1996) has been widely used in simulations to reduce variance in the estimation of the mean difference of two different random variables. In Cao (2007), the application of coupling methods in average reward performance potential was presented from the perspective of numerical solutions. This study found that coupling methods cannot improve the convergence rate of the numerical algorithms. Four coupling methods were presented in perturbation analysis to estimate the performance derivative in Dai (2000). This work pointed out that finding a better coupling scheme is desirable. The maximal coupling in Griffeath (1975) is maximal in the sense that merging is attained as efficiently as possible but leads to non-Markov coupling, and thus, the linear equation required by EGVR cannot be obtained. The basic idea regarding the use of coupling to estimate performance potential was briefly introduced in Li (2012) but details were not provided.

In this study, we present novel approaches for the EGVR of average reward performance potential based on the sample paths of Markov chains, which avoid the estimations of additional variables, such as average reward and steady state probability distribution, and thus implement EGVR in the true sense. On the basis of estimation algorithms, we introduce the correlation of the simulations of Markov chains by using coupling methods. Various coupling methods can be used to further reduce estimation variance or simulation time. Although certain coupling methods require information of transition probabilities, these methods can be applied to states where transition probabilities are known, to introduce the correlation of simulations or to combine them with physical simulations. In particular, we propose two optimal coupling methods that can minimize simulation time and estimation variance, respectively. To our knowledge, the simulation of Markov chains is frequently time-consuming and costly. The reduction in simulation time may significantly reduce estimation time. The coupling-based estimation methods used in this study can reduce estimation variance and simulation time and improve simulation efficiency.

2. Performance potential and its estimations

Consider an ergodic (irreducible, aperiodic, and positive recurrent) Markov chain $\boldsymbol{X} = \{X_l, l = 0, 1, \ldots\}$ on a finite state space $S = \{1, 2, \ldots, M\}$ with a transition probability matrix $P = [p(i,j)]_{i,j=1}^{M}$. Let $\pi = (\pi(1), \pi(2), \ldots, \pi(M))$ be a row vector that represents its steady state probability. Then, we derive the following balance equations:

$$\pi P = \pi, \pi e_M = 1, \tag{1}$$

where $e_M = (1, 1, \ldots, 1)^T$ is an M-dimensional column vector whose components are all equal to 1, and superscript "T" denotes the transpose. Let $f : S \to R$ be a reward function and occasionally a (column) reward vector, i.e., $f = (f(1), f(2), \ldots, f(M))^T$. We consider the long-run average reward (or simply the average reward) as a performance measure, which is defined as

$$\eta = \lim_{L\to\infty} \frac{1}{L} E\left\{\sum_{l=0}^{L-1} f(X_l)\right\} = \sum_{i=1}^{M} \pi(i) f(i) = \pi f.$$

For the aforementioned Markov chain, the following Poisson equation holds (Bertsekas, 1995; Cao, 2007; Puterman, 1994):

$$(I_M - P)g + \eta e_M = f, \tag{2}$$

where I_M denotes an $M \times M$ identity matrix. Its solution $g = (g(1), g(2), \ldots, g(M))^T$ is called average reward performance potential (Cao, 2007) (it is equivalent to the "relative cost function" (Bertsekas, 1995), "bias" (Puterman, 1994) or "value function" (Munos, 2006)). The solution to (2) can be obtained only up to an additive constant, i.e., if g is a solution to (2), then so is $g + ce_M$, where c is a constant. It can be proven that

$$g(i) = E\left\{\sum_{l=0}^{\infty} [f(X_l) - \eta] \middle| X_0 = i\right\}, i \in S \tag{3}$$

is a specific average-reward performance potential since

$$\begin{aligned} g(i) &= E\left\{\sum_{l=0}^{\infty} [f(X_l) - \eta] \middle| X_0 = i\right\} \\ &= f(i) - \eta + \sum_{j\in S} p(i,j) E\left\{\sum_{l=1}^{\infty} [f(X_l) - \eta] \middle| X_1 = j\right\} \\ &= f(i) - \eta + \sum_{j\in S} p(i,j) g(j), \end{aligned}$$

whose matrix form is the same as Eq. (2). To clearly describe EGVR, we also introduce the discounted reward performance potential, which is defined as follows:

$$g_\alpha(i) = E\left\{\sum_{l=0}^{\infty} \alpha^l f(X_l) \middle| X_0 = i\right\}, \tag{4}$$

where $0 < \alpha < 1$ denotes a discount factor. Similarly, the discounted reward performance potential satisfies the following Poisson equation (Puterman, 1994):

$$(I_M - \alpha P) g_\alpha = f, \tag{5}$$

where $g_\alpha = (g_\alpha(1), g_\alpha(2), \ldots, g_\alpha(M))^T$.

When the transition probabilities in P are known, performance potential g may be obtained by solving linear equation (2) or using its value iteration. However, certain transition probabilities in P may be generally unknown. In such case, the linear equation or its value iteration methods will not works. Estimation approaches can be applied for such cases. Monte Carlo algorithms are generally used to estimate performance potential and implement simulation-based optimization (Cooper et al., 2003; Marbach & Tsitsiklis, 2001). Assume that N samples of performance potential are obtained via simulation. Then, we have N estimates $\hat{g}^n(i)$, $n = 1, 2, \ldots, N$ of performance potential $g(i)$, $i \in S$. The final estimate is obtained by $\sum_{n=1}^{N} \hat{g}^n(i)/N$. The variance of this estimate is $\frac{1}{N^2}\sum_{n=1}^{N} var\hat{g}^n(i)$, which is reduced with order $1/N$. Compared with Monte Carlo algorithms, the EGVR has a geometric variance reduction rate. The related results about EGVR (Munos, 2006) are reviewed in the succeeding paragraphs.

Consider a function vector $g = (g(1), \ldots, g(M))^T$, where $g(i) = E[\Psi_i(f)]$ and $\Psi_i(f)$ is a linear sample path functional of reward function f that depends on the sample path of a Markov chain with initial state i. For example, discounted reward performance potential (4) can be described as $g_\alpha(i) = E[\Psi_i(f)]$ and

$$\Psi_i(f) = \sum_{l=0}^{\infty} \alpha^l f(X_l),$$

with $X_0 = i$. If function vector g satisfies an invertible linear equation $\mathcal{L}g = f$, where $\mathcal{L}$ is an invertible linear operator, then we have $g(i) = [\mathcal{L}^{-1}f](i) = E[\Psi_i(f)]$. If $\hat{g}$ is an estimate of g, then we have

$$g = \mathcal{L}^{-1}f = \hat{g} + \mathcal{L}^{-1}(f - \mathcal{L}\hat{g}) = \hat{g} + E[\Psi_i(f - \mathcal{L}\hat{g})]. \tag{6}$$

Therefore, the estimate of function $g(i)$ can be updated by the following iteration (Munos, 2006):

$$\begin{aligned}\hat{g}_{n+1}(i) &= \hat{g}_n(i) + E[\Psi_i(f - \mathcal{L}\hat{g}_n)] \\ &\approx \hat{g}_n(i) + \frac{1}{K}\sum_{k=1}^{K}\Psi_i^k(f - \mathcal{L}\hat{g}_n),\end{aligned} \tag{7}$$

where K sample paths are used to estimate the expectation $E[\Psi_i(f - \mathcal{L}\hat{g}_n)]$, and $\Psi_i^k(\cdot)$ denotes the functional $\Psi_i(\cdot)$ of the kth sample path. In (7), if $\hat{g}_n$ is a good approximation of g, then residual $f - \mathcal{L}\hat{g}_n$ is small and the estimation variance of $E[\Psi_i(f - \mathcal{L}\hat{g}_n)]$ is low. Let $v_n = \max_{i=1}^{S} var(\hat{g}_n(i))$. As shown in Munos (2006), the variance sequence $\{v_n\}$ of the aforementioned estimation satisfies,

$$v_{n+1} \le \rho_K v_n,$$

where $0 < \rho_K < 1$ when K is sufficiently large. Thus, variance can be reduced with geometric rate ρ_K.

Discounted reward performance potential (4) satisfies linear equation (5). Naturally, linear operator $\mathcal{L}$ is equal to $I_M - \alpha P$, which is invertible, and functional $\Psi_i(f)$ is $\sum_{l=0}^{\infty}\alpha^l f(X_l)$ with $X_0 = i$. Moreover, functional $\Psi_i(f)$ can be approximated by $\sum_{l=0}^{L}\alpha^l f(X_l)$, when L is sufficiently large, because α^l tends to zero when $l \to \infty$. However, for average reward performance potential, the case is more subtle and would deserve deeper treatment (Munos, 2006). An EGVR for an average reward case was presented in Munos (2006), in which Poisson equation (2) was used as a linear equation that average reward performance potential satisfies. Matrix $I_M - P$ in Poisson equation (2) is not invertible ($I_M - P$ has a zero eigenvalue since $Pe_M = e_M$), hence, a special performance potential with $\pi g = 0$ was considered and Eq. (2) was transformed into the following form (Munos, 2006):

$$\mathcal{L}g := (I_M - P + e_M\pi)g = f - \eta e_M. \tag{8}$$

Performance potential g that satisfies $\pi g = 0$ is the potential defined in (3). The functional

$$\Psi_i(f) = \sum_{l=0}^{\infty}[f(X_l) - \eta], \tag{9}$$

with $X_0 = i$ was used in Munos (2006) to estimate potential $g(i), i \in S$, such that $\pi g = 0$. However, this method will result in two problems. On the one hand, linear equation (8) contains average reward η and steady state probability π, and functional $\Psi_i(f)$ in (9) also contains average reward η. Consequently, additional variables η and π should be estimated while estimating potential g. In particular, average reward η should be re-estimated for different rewards $f - \mathcal{L}\hat{g}_n$ in each iteration (7). The estimations of these variables not only increase computation cost, but also the variance of performance potential estimates. On the other hand, functional $\Psi_i(f)$ in (9) is an infinite series, and thus a truncation parameter L^* should be introduced to approximate the functional by

$$\sum_{l=0}^{L^*}[f(X_l) - \eta], \tag{10}$$

with $X_0 = i$. This artificial truncation may result in a new error that will eliminate the benefit of EGVR. The selection of L^* is another difficulty. To overcome the aforementioned problems, we consider an improved EGVR for average reward performance potential based on PRF.

3. Improved estimation approach

Let $\boldsymbol{X} = \{X_l\}$ and $\mathbf{Y} = \{Y_l\}, l = 0, 1, 2, \ldots,$ be two Markov chains that start from state i and state j, respectively, and they have the same transition probability matrix P. PRF is defined as (Cao, 2007):

$$d(i,j) = E\left\{\sum_{l=0}^{L_{ij}^*-1}[f(Y_l) - f(X_l)]|X_0 = i, Y_0 = j\right\}, \tag{11}$$

where $L_{ij}^* = \min\{l : l \ge 0, Y_l = X_l|X_0 = i, Y_0 = j\}$ denotes the first time when two sample paths merge. Then, PRF satisfies (Cao, 2007)

$$d(i,i) = 0 \quad \text{and} \quad d(i,j) = -d(j,i), \quad i,j \in S, \tag{12}$$

and

$$d(i,j) = d(i,k) + d(k,j), \quad i,j,k \in S.$$

PRF quantifies the effect of a single perturbation from state i to state j and extends perturbation analysis theory to Markov systems (Cao, 2007). The perturbation realization matrix is defined as $D = [d(i,j)]_{i,j=1}^{M}$, and matrix D satisfies (Cao, 2007)

$$D - PDP^T = F, \tag{13}$$

with $D^T = -D$, where $F = e_M f^T - f e_M^T = [\widetilde{f}(i,j)]_{i,j=1}^{M}, \widetilde{f}(i,j) = f(j) - f(i)$. The relation between PRF and performance potentials is as follows (Cao, 2007):

$$d(i,j) = g(j) - g(i). \tag{14}$$

Performance potential is up to a constant, and thus we may fix one reference state i^* and set $g(i^*) = 0$. When PRFs are obtained, performance potentials $g(j), j \ne i^*$, are equal to $d(i^*, j)$. Thus, average reward performance potential can be estimated using PRFs (Cao, 2007).

Consider an operator $vec(\cdot)$ defined as follows (Abadir and Magnus, 2005):

$$\begin{aligned}&vec\left(\begin{bmatrix} a(1,1) & \cdots & a(1,n) \\ \vdots & \ddots & \vdots \\ a(n,1) & \cdots & a(n,n)\end{bmatrix}\right) \\ &= (a(1,1), \ldots, a(n,1), \ldots, a(1,n), \ldots, a(n,n))^T.\end{aligned}$$

Then, there is the following useful equation for this operator:

$$vec(ABC) = (C^T \otimes A)vec(B),$$

where A, B, and C are appropriate dimensional matrices and $\otimes$ denotes the Kronecker product. When this operator is used in Eq. (13), we obtain

$$(I_{M^2} - P \otimes P)vec(D) = vec(F). \tag{15}$$

Evidently, matrix $P \otimes P$ is a transition probability matrix of a Markov chain with state space $S \times S = \{(1,1), (1,2), \ldots, (1,M), (2,1), \ldots, (2,M), \ldots, (M,1), \cdots, (M,M)\}$, where $\times$ denotes the Cartesian product. Given that $D^T = -D$ and $F^T = -F$, Eq. (15) can be rewritten as

$$(I_{M^2} - P \otimes P)vec(D^T) = vec(F^T). \tag{16}$$

where $vec(D^T)$ and $vec(F^T)$ have the same order as state space $S \times S$. That is, $vec(D^T) = (d(1,1), d(1,2), \ldots, d(1,M), \ldots, d(M,1), \ldots, d(M,M))^T$ and $vec(F^T) = (\widetilde{f}(1,1), \widetilde{f}(1,2), \ldots, \widetilde{f}(1,M), \ldots, \widetilde{f}(M,1), \ldots, \widetilde{f}(M,M))^T$.

Compared with Eqs. (2), (16) does not include average reward and steady state distribution. We intend to use this linear equation

and iteration (7) to obtain the EGVR of PRF, i.e., the estimation of performance potential. However, matrix $I_{M^2} - P \otimes P$ is also a singular matrix, and thus, the solution for Eq. (16) is not unique, which appears to prevent the application of iteration (7). Subsequently, we show that PRF can be determined uniquely by Eq. (16) under conditions (12), and thus, the inverse $\mathcal{L}^{-1}$ of Eq. (16) still exist under conditions (12). We introduce the following lemma from (Berman and Plemmons 1994, p. 156, Theorem 4.16).

Lemma 1. *We assume that $B \geq 0$ is a non-negative matrix and $s > 0$ satisfies $s \geq \rho(B)$, which is the spectral radius of matrix B. Let matrix $A = sI - B$, which is called M-matrix in Berman and Plemmons (1994), be a singular and irreducible matrix. Each principal submatrix of A apart from A itself is a nonsingular M-matrix.*

Theorem 1. *The solution that satisfies $d(i, i) = 0$ and $d(i, j) = -d(j, i)$, $i, j = 1, 2, \ldots, M$ to (16) is unique.*

Proof. We divide the state space $S \times S$ into three subsets: $S_1 = \{(1, 1), \ldots, (M, M)\}$ denotes the diagonal state set, $S_2 = \{(1, 2), (1, 3), \ldots, (M-1, M)\}$ denotes the upper triangular state set, and $S_3 = \{(2, 1), (3, 1), \ldots, (M, M-1)\}$ denotes the lower triangular state set. Under this division, we obtain the following equation by rearranging the order of state space:

$$vec(D^T) = \left(\mathbf{d}_{S_1}^T\ \mathbf{d}_{S_2}^T\ \mathbf{d}_{S_3}^T\right)^T = \left(\mathbf{0}^T\ \mathbf{d}^T\ -\mathbf{d}^T\right)^T,$$

where $\mathbf{0}$ is an M-dimensional column vector whose elements are all equal to 0 and $\mathbf{d} = (d(1, 2), d(1, 3), \ldots, d(M-1, M))^T$,

$$vec(F^T) = \left(\mathbf{f}_{S_1}^T\ \mathbf{f}_{S_2}^T\ \mathbf{f}_{S_3}^T\right)^T = \left(\mathbf{0}^T\ \mathbf{f}^T\ -\mathbf{f}^T\right)^T,$$

where $\mathbf{f} = (\widetilde{f}(1, 2), \widetilde{f}(1, 3), \ldots, \widetilde{f}(M-1, M))^T$, and

$$I_{M^2} - P \otimes P = I_{M^2} - \begin{bmatrix} P_{S_1S_1} & P_{S_1S_2} & P_{S_1S_3} \\ P_{S_2S_1} & P_{S_2S_2} & P_{S_2S_3} \\ P_{S_3S_1} & P_{S_3S_2} & P_{S_3S_3} \end{bmatrix}.$$

In accordance with the property of the Kronecker product, we obtain

$$P_{S_1S_2} = P_{S_1S_3}, P_{S_2S_2} = P_{S_3S_3}, P_{S_2S_3} = P_{S_3S_2}.$$

On the basis of the aforementioned divisions, Eq. (16) is simplified as

$$(I_{M(M-1)/2} - P_{S_2S_2} + P_{S_2S_3})\mathbf{d} = \mathbf{f}. \tag{17}$$

From the ergodic property of Markov chain $\boldsymbol{X}$, matrix $P \otimes P$ is an irreducible matrix. Given that $\rho(P \otimes P) = 1$, $I_{M^2} - P \otimes P$ is a singular and irreducible M-matrix. $I_{M(M-1)/2} - P_{S_2S_2}$ is a principal submatrix of matrix $I_{M^2} - P \otimes P$, and thus, $I_{M(M-1)/2} - P_{S_2S_2}$ is nonsingular. Then matrix $I_{M(M-1)/2} - P_{S_2S_2} + P_{S_2S_3}$ is invertible and its inversion can be obtained using the matrix inversion lemma. Therefore, the solution for (17) is unique and so is the solution that satisfies $d(i, i) = 0$ and $d(i, j) = -d(j, i)$, $i, j = 1, 2, \ldots, M$ to (16). □

Theorem 1 guarantees the uniqueness of the solution for linear equation (16) under constraints (12). PRF vector $vec(D^T)$ is a solution for Equation (16) and satisfies constraints (12). Thus, PRF can be determined uniquely via Eq. (16) under conditions (12). Then, by using the composite process $\{(X_l, Y_l), l = 0, 1, \ldots\}$ of Markov chains $\boldsymbol{X}$ and $\mathbf{Y}$, which has the transition probability matrix $P \otimes P$, and setting $\mathcal{L} = I - P \otimes P$, $f = vec(F^T)$ and $\Psi_{(i,j)} = \sum_{l=0}^{L_{ij}^*-1}[f(Y_l) - f(X_l)]$ with $(X_0, Y_0) = (i, j)$, iteration (7) can be applied to estimate PRF $d(i, j)$ if we fix $d(i, i) = 0$ and $d(i, j) = -d(j, i)$ for all $i, j \in S$ during the iteration process.

Estimating average reward performance potentials via PRFs has two benefits. On one hand, linear equation (16) uniquely determines PRFs and does not contain steady state probability and average reward. Thus, it can be directly applied to EGVR without estimating average reward and steady state probability. On the other hand, the sample path functional

$$\sum_{l=0}^{L_{ij}^*-1}[f(Y_l) - f(X_l)]$$

in PRF provides natural termination, and thus, artificial truncation in (10) is avoided.

4. Coupling-based estimations

To our knowledge, correlation can be applied to reduce variance (Whitt, 1976). For example, we estimate $E(X - Y)$ via simulation, where X and Y are two random variables with probability distribution functions F and G, respectively. We generate random variables X and Y using the inverse transformation method, which means $X = F^{-1}(\xi_1)$ and $Y = G^{-1}(\xi_2)$. From Whitt (1976), we determine that $Var(X - Y)$ is the smallest among all possible ξ_1 and ξ_2 if $\xi_1 = \xi_2$. The improved approach presented in Section 3 does not introduce any correlation between $\boldsymbol{X}$ and $\mathbf{Y}$ because the transition matrix $P \otimes P$ is only their independent combination. To estimate PRFs, two Markov chains do not require to be independent (Cao, 2007). In this section, we use the coupling methods in Griffeath (1975), Lindvall (1992) and Pflug (1996) to introduce a correlation between two Markov chains and further improve simulation efficiency.

Consider a composite Markov chain $\widetilde{\boldsymbol{X}} = \{(\widetilde{X}_l, \widetilde{Y}_l), l = 0, 1, 2 \ldots\}$ with state space $S \times S$. Its transition probability matrix is $\widetilde{P} = \left[\widetilde{p}[(i, j), (r, s)]\right]_{i,j,r,s \in S}$, where

$$\widetilde{p}[(i, j), (r, s)] = \mathcal{P}\left(\widetilde{X}_{l+1} = r, \widetilde{Y}_{l+1} = s | \widetilde{X}_l = i, \widetilde{Y}_l = j\right).$$

If the marginal probability distributions of $\widetilde{\boldsymbol{X}}$ with respect to $\widetilde{X}_l$ and $\widetilde{Y}_l$ are the same as the distributions of $\boldsymbol{X}$ and $\mathbf{Y}$, respectively, then Markov chain $\widetilde{\boldsymbol{X}}$ is a coupling of Markov chains $\boldsymbol{X}$ and $\mathbf{Y}$ (Cao, 2007). Thus, if

$$\sum_{s=1}^{M} \widetilde{p}[(i, j), (r, s)] = p(i, r), \quad \forall \quad i, j, r \in S, \tag{18}$$

$$\sum_{r=1}^{M} \widetilde{p}[(i, j), (r, s)] = p(j, s), \quad \forall \quad i, j, s \in S, \tag{19}$$

then Markov chain $\widetilde{\boldsymbol{X}}$ is a coupling of Markov chains $\boldsymbol{X}$ and $\mathbf{Y}$.

For composite Markov chain $\widetilde{\boldsymbol{X}}$, a reward function $\widetilde{f}(i, j) = f(j) - f(i)$, $i, j \in S$, is defined and let $\widetilde{f}$ be its column vector form, i.e., $\widetilde{f} = vec(F^T)$. Then, the following lemma provides the expression for the average reward of $\widetilde{\boldsymbol{X}}$. A related expression was established in Cao (2007). We provide a concise self-contained proof given the importance of this result.

Lemma 2. *The steady state probability distribution $\widetilde{\pi}$ of Markov chain $\widetilde{\boldsymbol{X}}$ satisfies $\sum_{j \in S} \widetilde{\pi}(i, j) = \pi(i)$ and $\sum_{i \in S} \widetilde{\pi}(i, j) = \pi(j)$. Thus, we have $\widetilde{\eta} = \widetilde{\pi}\widetilde{f} = 0$.*

Proof. Given that the steady state probability distribution $\widetilde{\pi}$ of Markov chain $\widetilde{\boldsymbol{X}}$ satisfies the balance equation, $\widetilde{\pi}\widetilde{P} = \widetilde{\pi}$ and $\widetilde{\pi} e_{M^2} = 1$, than we obtain

$$\sum_{i,j \in S} \widetilde{\pi}(i, j)\widetilde{p}[(i, j), (r, s)] = \widetilde{\pi}(r, s), \forall r, s \in S,$$

$$\sum_{i \in S} \sum_{j \in S} \widetilde{\pi}(i, j) = 1. \tag{20}$$

When both sides of (20) are summed up for all $s \in S$ and using (18), we derive

$$\sum_{i\in S}\sum_{j\in S}\widetilde{\pi}(i,j)p(i,r) = \sum_{s\in S}\widetilde{\pi}(r,s).$$

From the uniqueness of balance equations $\pi P = \pi$ and $\pi e_M = 1$, we obtain $\sum_{j\in S}\widetilde{\pi}(i,j) = \pi(i)$. Similarly, we may prove $\sum_{i\in S}\widetilde{\pi}(i,j) = \pi(j)$. Then,

$$\begin{aligned}\widetilde{\eta} &= \sum_{i,j\in S}\widetilde{\pi}(i,j)\widetilde{f}(i,j) = \sum_{i\in S}\sum_{j\in S}\widetilde{\pi}(i,j)[f(j)-f(i)]\\ &= \sum_{j\in S}\pi(j)f(j) - \sum_{i\in S}\pi(i)f(i) = \eta - \eta = 0. \quad \square\end{aligned}$$

Similar to Poisson equation (2), the following Poisson equation for composite Markov chain $\widetilde{\boldsymbol{X}}$ is derived given that $\widetilde{\eta} = 0$:

$$(I_{M^2} - \widetilde{P})\widetilde{g} = \widetilde{f}, \tag{21}$$

where $\widetilde{g}$ is the average reward performance potential that corresponds to chain $\widetilde{\boldsymbol{X}}$, and similarly to performance potential (3),

$$\widetilde{g}(i,j) = E\left\{\sum_{l=0}^{\infty}\left[f(\widetilde{Y}_l) - f(\widetilde{X}_l)\right]\Big|(\widetilde{X}_0,\widetilde{Y}_0) = (i,j)\right\}, \tag{22}$$

for all $(i,j) \in S \times S$, is a specific performance potential for chain $\widetilde{\boldsymbol{X}}$.

From the definition of the Kronecker product, matrix $P \otimes P$ in Section 3 satisfies the coupling conditions (18) and (19). We call this Kronecker coupling (KC). KC guarantees the existence of coupled matrix $\widetilde{P}$. However it is only the independent combination of Markov chains $\boldsymbol{X}$ and $\mathbf{Y}$ and does not introduce any correlation between $\boldsymbol{X}$ and $\mathbf{Y}$. Under KC, and considering the strong Markov property, two independent Markov chains behave similarly in the statistical sense after L^*_{ij}; therefore, $\widetilde{g}(i,j)$ in (22) is equal to $d(i,j)$ and Eq. (21) is the same as Eq. (16). For other coupling methods, however, the results under KC cannot be directly extended. The following several questions should be considered.

(1) Does performance potential $\widetilde{g}(i,j)$ have the same functional form as (11) under different coupling methods? Can PRF be determined uniquely using Eq. (21) under different coupling methods?
(2) What is the optimal $\widetilde{P}$ for enhancing the simulation efficiency?

The second question is answered in Section 5. To answer the first question, the structural property of the transition probability matrices of different coupling methods should be considered. The proof of Theorem 1 depends on the irreducible property of the KC matrix, which does not apply to other coupling methods.

First, Doeblin coupling (Lindvall, 1992) is a classical coupling method that is obtained by setting $\widetilde{p}[(i,i),(s,s)] = p(i,s)$, $\widetilde{p}[(i,i),(r,s)] = 0, r \neq s \in S$ for all $i \in S$ and $\widetilde{p}[(i,j),(r,s)] = p(i,r)p(j,s)$ for all $i \neq j \in S$. Evidently, the transition matrix of Doeblin coupling satisfies coupling conditions. If the state space is arranged by order $\left\{S_1, S_2, S_3\right\}$, where S_1, S_2, and S_3 are the state sets defined in the proof of Theorem 1, then the coupled transition probability matrix can be described by blocks as follows:

$$\widetilde{P} = \begin{bmatrix} P & \mathbf{0} \\ Q & R \end{bmatrix}, \tag{23}$$

where Q and R are matrices with suitable dimensions. This transition probability matrix is a reducible matrix, and thus, Theorem 1 cannot be directly applied to Doeblin coupling. Under the matrix structure of (23), coupling time L^*_{ij} is the time that the process moves to set $\{(i,i), i \in S\}$ from state (i,j). Its mean can be computed by the corresponding row of $(I_{M^2-M} - R)^{-1}e_{M^2-M}$.

Second, the Common Random Numbers (CRN) method (Pflug, 1996), which can be easily implemented, is considered and a positive correlation between two Markov chains is introduced by simulating two chains with a common sequence of random numbers. Under the CRN coupling method, transition probabilities $\widetilde{p}[(i,i),(s,s)] = p(i,s), s \in S$ and $\widetilde{p}[(i,i),(r,s)] = 0, r \neq s \in S$ for all $i \in S$. Thus, the composite transition matrix can also be described by (23). Transition probabilities $\widetilde{p}[(i,j),(r,s)]$ for the $i \neq j$ case, i.e., the probabilities in Q and R in (23), can be obtained as follows by comparing the cumulative probabilities starting from states i and j.

(1) Cumulative transition probabilities $p_i(t) = \sum_{r=1}^{t}p(i,r)$ and $p_j(t) = \sum_{s=1}^{t}p(j,s)$, $t = 1,\ldots,M$ are computed.
(2) Probabilities $\{p_i(1),\ldots,p_i(M),p_j(1),\ldots,p_j(M)\} =: \{A(1),\ldots,A(2M)\}$ are sorted in ascending order. Let $\{a_1,a_2,\ldots,a_{2M}\}$ denote the index in ascending order, i.e., $A(a_1) \leq A(a_2) \leq \cdots \leq A(a_{2M})$.
(3) Set $r = 1, s = 1, \widetilde{p}[(i,j),(1,1)] = A(a_1)$. Let m increase from 1 to $\mathcal{M}$, where $\mathcal{M}$ satisfies $A(a_{\mathcal{M}}) = 1$, if $a_m > M, s := s+1$; otherwise, $r := r+1$, set $\widetilde{p}[(i,j),(r,s)] = A(a_{m+1}) - A(a_m)$. Other one-step transition probabilities from (i,j) are equal to zero.

The following example is provided to demonstrate the computation of transition probability $\widetilde{p}[(2,3),(r,s)]$ for $r,s \in S$. Let $p(2,1) = 0.5, p(2,2) = 0.3, p(2,3) = 0.2$ and $p(3,1) = 0.2$, $p(3,2) = 0.7, p(3,3) = 0.1$. Then, $(p_2(1),p_2(2),p_2(3)) = (A(1),A(2),A(3)) = (0.5,0.8,1)$ and $(p_3(1),p_3(2),p_3(3)) = (A(4),A(5),A(6)) = (0.2,0.9,1)$. These value can be sorted as $(p_3(1),p_2(1),p_2(2),p_3(2),p_2(3),p_3(3)) = (A(4),A(1),A(2),A(5),A(3),A(6))$. From step (3), we derive $p[(2,3),(1,1)] = A(4) = 0.2$. Since $a_1 = 4 > M = 3$, we obtain $s := s+1 = 2$, and thus, we set $p[(2,3),(1,2)] = A(1) - A(4) = 0.3$. Similarly, we can obtain $p[(2,3),(2,2)] = 0.3, p[(2,3),(3,2)] = 0.1$, and $p[(2,3),(3,3)] = 0.1$. Other transition probabilities from $(2,3)$ are equal to zero.

Third, from the definition of PRF, coupling time L^*_{ij} and the number of simulation steps will decrease if two Markov chains are merged as soon as possible. We consider a coupling to merge two Markov chains as soon as possible in one step, that is, we intend to maximize one-step transition probability $\widetilde{p}[(i,j),(s,s)]$ for all $i \neq j$ and $s \in S$. Given that

$$\widetilde{p}[(i,j),(s,s)] \leq \min(p(i,s),p(j,s)) =: p(i,s) \wedge p(j,s).$$

If $\widetilde{p}[(i,j),(s,s)] = p(i,s) \wedge p(j,s)$, than the maximal one-step coupling probability can be obtained. Vasershtein coupling (VC) (Griffeath, 1975) provides this type of coupling, the coupled matrix of which is defined as

$$\begin{aligned}&\widetilde{p}[(i,i),(s,s)] = p(i,s),\\ &\widetilde{p}[(i,j),(s,s)] = p(i,s) \wedge p(j,s),\\ &\widetilde{p}[(i,j),(r,s)]\\ &= \frac{\left(p(i,r)-p(j,r)\right)^+\left(p(j,s)-p(i,s)\right)^+}{1-\delta_{i,j}}, \quad r \neq s,\end{aligned} \tag{24}$$

where $(\cdot)^+$ denotes $\max(\cdot,0)$ and $\delta_{i,j} = \sum_{s\in S}p(i,s) \wedge p(j,s)$. Thus, the transition matrix has the same structure as (23).

The three aforementioned coupling methods have the same structure as (23) in their transition probability matrices. Under such structure, state set S_1 is an absorbing set. L^*_{ij} in (11) is the time when the process first enters S_1. Two Markov chains have the same sample path after entering S_1, and thus, performance potential (22) can be rewritten as

$$\widetilde{g}(i,j) = E\left\{\sum_{l=0}^{L^*_{ij}-1}\left[f(\widetilde{Y}_l) - f(\widetilde{X}_l)\right]\Big|(\widetilde{X}_0,\widetilde{Y}_0) = (i,j)\right\}. \tag{25}$$

Therefore, performance potential $\widetilde{g}(i,j)$ has the same functional form as PRF $d(i,j)$ in (11).

In Doeblin coupling, Q and R are the same as those in KC. Thus, $\widetilde{g}(i,j)$ in (25) is equal to $d(i,j)$. Similar to KC, Doeblin coupling does not introduce any correlation and evolves independently before entering set S_1. Thus, Doeblin coupling is entirely the same as KC in estimating average reward performance potential. The difference between Doeblin coupling and KC is that the former introduces a correlation after entering S_1, whereas KC never introduces a correlation. Subsequently, we prove that $\widetilde{g}(i,j)$ in (25) is equal to $d(i,j)$ for the other aforementioned coupling methods.

First, we introduce the following lemma from Cao (2007, p.160).

Lemma 3. *PRF vector $vec(D^T) = e_M \otimes g - g \otimes e_M$ is a solution for linear equation (21) for all coupled transition matrices $\widetilde{P}$.*

Theorem 2. *Under couplings with matrix structure (23), performance potential $\widetilde{g}(i,j)$ in (25) is equal to $d(i,j)$ and can be determined uniquely by Eq. (21).*

Proof. To simplify the notation, let $S_1^c = S_2 \bigcup S_3$. We have $\widetilde{g}(i,i) = 0$ for all $(i,i) \in S_1$ because $\widetilde{X}_l$ and $\widetilde{Y}_l$, $l = 0, 1, \ldots,$ which start from $(i,i) \in S_1$, have the same transitions. Moreover, matrix $\widetilde{P}$ has the same form as (23). Thus, Eq. (21) can be written as

$$\begin{bmatrix} I_M - P & \mathbf{0} \\ -Q & I_{M^2-M} - R \end{bmatrix} \begin{bmatrix} \mathbf{0} \\ \widetilde{g}_{S_1^c} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \widetilde{f}_{S_1^c} \end{bmatrix},$$

where $\widetilde{g}_{S_1^c}$ and $\widetilde{f}_{S_1^c}$ denote the restrictions of $\widetilde{g}$ and $\widetilde{f}$ in set S_1^c, respectively. The transition probability matrix (23) is a canonical form, and thus, we have $\rho(R) < 1$ and $I_{M^2-M} - R$ is invertible (see Lemma B.2 in Cao (2007)). Therefore, the solution for (21) is unique. For composite Markov chain $\widetilde{\boldsymbol{X}}$, it is easy to demonstrate that performance potential $\widetilde{g}$ in (22) or (25) is a solution for linear equation (21) with $\widetilde{g}(i,i) = 0, i \in S$. From Lemma 3, PRF vector $vec(D^T)$ is also a solution for Eq. (21) with $\widetilde{g}(i,i) = 0$. Therefore, with regard to uniqueness, $\widetilde{g}$ in (22) or (25) is equal to $vec(D^T)$ and they are unique solution for (21) with $\widetilde{g}(i,i) = 0, i \in S$. □

Theorem 2 guarantees the existence of an inverse of the linear operator in (21) if we fix $\widetilde{g}(i,i) = 0, i \in S$. Thus, linear equation (21) can be applied to iteration (7) to obtain the EGVR of PRFs if $\widetilde{g}(i,i) = 0$ for all $i \in S$ are fixed. That is, we may use the CRN and VC methods to simulate a composite Markov chain $\widetilde{\boldsymbol{X}}$ and use functional

$$\sum_{l=0}^{L_{ij}^*-1} \left[f(\widetilde{Y}_l) - f(\widetilde{X}_l) \right],$$

with an exact termination to estimate PRFs in (7).

5. Optimal coupling methods

In this section, we consider what is the optimal $\widetilde{P}$ for enhancing simulation efficiency and provide two types of new coupling methods.

We continue to consider VC. The sample paths of VC can be simulated in two steps. Let $\widetilde{X}_l = i$ and $\widetilde{Y}_l = j$ at time l. In the first step, a uniform distributed random variable $\xi_l \in [0,1)$ is generated. If $\sum_{s=1}^{t-1} p(i,s) \wedge p(j,s) \le \xi_l < \sum_{s=1}^{t} p(i,s) \wedge p(j,s)$, $t \in S$ (with the convention $\sum_{s=1}^{0} p(i,s) \wedge p(j,s) = 0$), then we set $\widetilde{X}_{l+1} = \widetilde{Y}_{l+1} = t$, which indicates that the process will merge at the same state t with the largest probability $p(i,t) \wedge p(j,t)$; otherwise, proceed to the second step. We generate two independent random variables U_l and V_l with distributions $p_U(s) = \frac{\left(p(i,s)-p(j,s)\right)^+}{1-\delta_{i,j}}$ and $p_V(s) = \frac{\left(p(j,s)-p(i,s)\right)^+}{1-\delta_{i,j}}$, $s \in S$. If $U_l = r$ and $V_l = s$, then we set $\widetilde{X}_{l+1} = r$ and $\widetilde{Y}_{l+1} = s$. From the simulation process, the transition probability from state (i,j) to (r,s), $r \ne s$ is $(1-\delta_{i,j})\frac{\left(p(i,r)-p(j,r)\right)^+\left(p(j,s)-p(i,s)\right)^+}{(1-\delta_{i,j})^2} = \frac{\left(p(i,r)-p(j,r)\right)^+\left(p(j,s)-p(i,s)\right)^+}{1-\delta_{i,j}}$. Thus, the transition probabilities under the preceding simulation are equal to (24). The second step determines the transition probabilities from state (i,j), $i \ne j$ to state (r,s), $r \ne s$, i.e., the probabilities in R in matrix (23), by two independent random variables. Evidently, there are still available space to introduce the coupling correlation and improve the mergence. On the basis of this idea, we intend to find new transition probabilities in R and Q to improve simulation efficiency. We expect the maximal coupling in Griffeath (1975), however it leads to a non-Markov coupling, and thus, a linear equation cannot be obtained. Subsequently, we consider two new optimal coupling methods.

First, we present a coupling method by solving a constrained stochastic shortest problem (SSP) and call this method SSP coupling. To make the process merge as soon as possible, coupled transition probabilities in R should be found to minimize the path that starts from set S_1^c to set S_1. Simultaneously, the transition probabilities in R should satisfy coupling conditions (18) and (19). We define a new reward function $\bar{f}(i,j) = 1$ for $(i,j) \in S_1^c$ and transform the aforementioned problem into the following constrained SSP problem. A reward $\bar{f}(i,j)$ is incurred when the state is $(i,j) \in S_1^c$. Submatrix R is the transition probability matrix of the states in set S_1^c. Subset S_1 is an absorbing set, and the reward is 0 if the state is in S_1. If the state moves to set S_1, then the process stops. The objective of SSP is to find a transition matrix R to minimize the following total reward

$$J(i,j) = E\Big[\sum_{l=0}^{\infty} \bar{f}(\widetilde{X}_l, \widetilde{Y}_l) | (\widetilde{X}_0, \widetilde{Y}_0) = (i,j)\Big], \tag{26}$$

for all $(i,j) \in S_1^c$, which computes the expected total path from state $(i,j) \in S_1^c$ to set S_1. Simultaneously, the transition probabilities in R satisfy coupling conditions (18) and (19).

Let J and $\bar{f}$ denote the vector forms of total reward (26) and reward function $\bar{f}(i,j)$ for all $(i,j) \in S_1^c$, respectively. Then, we have (Bertsekas, 1995)

$$\bar{f} + RJ = J. \tag{27}$$

Let J' be another total reward under another transition matrix R'. Then, we have,

$$\bar{f} + R'J' = J'. \tag{28}$$

From (27) and (28), we derive

$$J' - J = (\bar{f} + R'J') - (\bar{f} + RJ) = R'(J' - J) + (R' - R)J.$$

Thus, the following performance difference can be obtained:

$$J' - J = (I_{M^2-M} - R')^{-1}(R' - R)J. \tag{29}$$

For two N-dimensional vectors u and v, $u \le v$ indicates $u(i) \le v(i)$ for all $i = 1, 2, \ldots, N$. If we choose R', such that $R'J \le RJ$, which indicates $(R' - R)J \le 0$, then we have $J' \le J$ from (29) because R' is a non-negative matrix and $(I_{M^2-M} - R')(R' - R)J \le 0$. Each row of R is independent, and thus, we may use the following linear programming to obtain each row of R', such that $R'J \le RJ$:

$$\begin{aligned} \min \quad & R_{ij} J \\ s.t. \quad & R_{ij} \text{ is a nonnegative row vector that} \\ & \text{satisfies conditions (18) and (19)} \end{aligned} \tag{30}$$

where R_{ij} denotes the transition probability distribution from state (i,j), i.e., the (i,j)th row of transition matrix R. From the preceding

analysis, we may design the following policy iteration type algorithm with linear programming (Algorithm 1) to obtain an optimal transition matrix R. Linear programming (31) is the realization of (30) for each row of R. Given that at least one zero in $p(i,r)-p(i,r)\wedge p(j,r)$ and $p(j,r)-p(i,r)\wedge p(j,r)$, $r=1,\ldots,M$, M nonzero equality constraints exist at most in (32) and (33), which lead to numerous zeros in R.

Algorithm 1

(1) Given an initial matrix R_0 (such as R in VC), set $n=0$ and $J_{-1}=0$.

(2) (Policy evaluation) Compute the value function using (27), i.e., $J_n=(I_{M^2-M}-R_n)^{-1}\widetilde{f}$. If $\|J_n-J_{n-1}\|\le\epsilon$, then return R_n and stop; otherwise, proceed to Step 3.

(3) (Policy improvement) Compute the improved transition probability distribution $R_{n+1}[(i,j),(r,s)]$, which is written as $R(r,s)$ for simplification, for each state (i,j), $i\neq j\in\mathcal{S}$, using the following linear programming problem:

$$\min\sum_{r,s=1,r\neq s}^{M}J_n(r,s)R(r,s), \tag{31}$$

$$s.t.\ \sum_{s=1,s\neq r}^{M}R(r,s)=p(i,r)-p(i,r)\wedge p(j,r), \tag{32}$$

$$\sum_{r=1,r\neq s}^{M}R(r,s)=p(j,s)-p(i,s)\wedge p(j,s), \tag{33}$$

$$R(r,s)\ge 0,\ r,s\in S.$$

(4) Record J_n, set $n:=n+1$, and return to Step 2.

The aforementioned SSP coupling can merge two Markov chains as soon as possible, and thus, the number of simulation steps can be reduced. However, minimizing mergence time is not equivalent to minimizing variance. To minimize variance of estimation, we present a method with minimal PRF variance by solving a variance minimization problem and call this method variance minimization (VM) coupling.

The variance of PRF is defined as

$$\nu(i,j)=var\Big\{\sum_{l=0}^{L_{ij}^*-1}[f(\widetilde{Y}_l)-f(\widetilde{X}_l)]|(\widetilde{X}_0,\widetilde{Y}_0)=(i,j)\Big\},$$

whereas the second moment of PRF is defined as

$$\omega(i,j)=E\Big\{\Big[\sum_{l=0}^{L_{ij}^*-1}[f(\widetilde{Y}_l)-f(\widetilde{X}_l)]\Big]^2|(\widetilde{X}_0,\widetilde{Y}_0)=(i,j)\Big\}.$$

From the variance formula, we derive

$$\nu(i,j)=\omega(i,j)-d(i,j)^2,\ (i,j)\in S\times S.$$

Evidently, $\omega(i,i)=0$ for all $i\in S$ under matrix structure (23). The PRF is fixed for all the coupled transition matrices, and thus, the variance is equivalent to the second moment. Accordingly, the coupling problem can be transformed to a second moment minimization problem, which denotes finding the coupled probabilities in Q and R to minimize second moment $\omega(i,j)$ for all $(i,j)\in S_1^c$ and simultaneously satisfy coupling conditions (18) and (19).

For the second moment of PRF, we derive

$$\begin{aligned}\omega(i,j)=&\widetilde{f}(i,j)^2+\\&2\widetilde{f}(i,j)E\Big\{\Big[\sum_{l=1}^{L_{ij}^*-1}[f(\widetilde{Y}_l)-f(\widetilde{X}_l)]\Big]|(\widetilde{X}_0,\widetilde{Y}_0)=(i,j)\Big\}\\&+E\Big\{\Big[\sum_{l=1}^{L_{ij}^*-1}[f(\widetilde{Y}_l)-f(\widetilde{X}_l)]\Big]^2|(\widetilde{X}_0,\widetilde{Y}_0)=(i,j)\Big\}\\=&\widetilde{f}(i,j)^2+2\widetilde{f}(i,j)\sum_{(r,s)\in S_1^c}\widetilde{p}[(i,j),(r,s)]d(r,s)\\&+\sum_{(r,s)\in S_1^c}\widetilde{p}[(i,j),(r,s)]\omega(r,s),\end{aligned}$$

where we have used $d(i,i)=0$ and $\omega(i,i)=0$ for all $i\in S$. Rewriting the preceding equation in vector form for all $(i,j)\in S_1^c$, we obtain

$$\omega=\widetilde{f}^2+2\widetilde{F}R\widetilde{g}+R\omega, \tag{34}$$

where we restrict vectors $\widetilde{f}$, $\widetilde{g}$, and ω in set S_1^c. $\widetilde{f}^2$ denotes a vector whose component is the square of each component of $\widetilde{f}$ and $\widetilde{F}=diag\{\widetilde{f}\}$.

Let ω' denote the second moment under another R'. Then, we have

$$\omega'=\widetilde{f}^2+2\widetilde{F}R'\widetilde{g}+R'\omega'. \tag{35}$$

From (34) and (35), we obtain

$$\begin{aligned}\omega'-\omega&=(\widetilde{f}^2+2\widetilde{F}R'\widetilde{g}+R'\omega')-(\widetilde{f}^2+2\widetilde{F}R\widetilde{g}+R\omega)\\&=2\widetilde{F}(R'-R)\widetilde{g}+R'(\omega'-\omega)+(R'-R)\omega.\end{aligned}$$

Thus, the following performance difference formula can be derived:

$$\omega'-\omega=(I_{M^2-M}-R')^{-1}[2\widetilde{F}(R'-R)\widetilde{g}+(R'-R)\omega]. \tag{36}$$

Similarly, from the non-negative property of R', if $2\widetilde{F}(R'-R)\widetilde{g}+(R'-R)\omega\le 0$, i.e., $2\widetilde{F}R'\widetilde{g}+R'\omega\le 2\widetilde{F}R\widetilde{g}+R\omega$, we have $\omega'\le\omega$. Evidently, $2\widetilde{F}R'\widetilde{g}+R'\omega$ is linear with respect to R'. We may obtain an improved Q' and R' by minimizing $2\widetilde{F}R'\widetilde{g}+R'\omega$ under coupling conditions (18) and (19). Thus, if PRFs $d(i,j)$, $i\neq j$, are given beforehand (in application, the estimates of PRFs can be used), we may also use a policy iteration type algorithm with linear programming (Algorithm 2) to obtain the optimal coupled transition probabilities in R and Q.

SSP coupling and VM coupling provide the optimal transition probabilities through additional calculations. Compared with long simulations, these additional calculations are more advantageous. In general, the simulation time of Markov chains is considerably longer than the computation time of linear programming. The additional computation time can be regarded as a "thinking" time to consider how the simulation should be performed. Of course, thinking time depends on the size of state space and linear programming. However, thinking time is less than the simulation time saved by the coupling methods; consequently, the total time is not increased. From the simulation results in the next section, we determine that these coupling methods can reduce estimation variance and simulation time and improve the efficiency of simulations.

Remark. (1) KC and Doeblin coupling are obtained from the product of transition probabilities of two original Markov chains. The simulation of a sample path under these two couplings can be directly implemented by combining two sample paths of original systems. (2) In CRN, VC, SSP coupling, and VM coupling, the computation of coupling probabilities at state (i,j) requires the

Algorithm 2

(1) Given an initial matrices Q_0 and R_0 (such as Q and R in CRN), set $n = 0$ and $\omega_{-1} = \mathbf{0}$.

(2) (Policy evaluation) Compute the second moment vector using (34), i.e., $\omega_n = (I_{M^2-M} - R_n)^{-1}[\widetilde{f}^2 + 2\widetilde{F}R_n\widetilde{g}]$. If $\|\omega_n - \omega_{n-1}\| \le \epsilon$, then return Q_n, R_n, and stop; otherwise, proceed to Step 3.

(3) (Policy improvement) Compute the improved transition probability distribution $R_{n+1}[(i,j),(r,s)]$ and $Q_{n+1}[(i,j),(r,r)]$, which are respectively written as $R(r,s)$ and $Q(r,r)$ for simplification, for each state (i,j), $i \neq j \in \mathcal{S}$, by solving the following linear programming:

$$\min \sum_{r,s=1}^{M} [2\widetilde{f}(i,j)d(r,s) + \omega_n(r,s)]R(r,s),$$

$$s.t. \sum_{s=1,s\neq r}^{M} R(r,s) + Q(r,r) = p(r|i),$$

$$\sum_{r=1,r\neq s}^{M} R(r,s) + Q(s,s) = p(s|j),$$

$$R(r,s) \ge 0,\ Q(r,r) \ge 0,\ Q(s,s) \ge 0,\ r,s \in S.$$

(4) Record ω_n, set $n := n+1$, and return to Step 2.

transition probabilities at original states i and j. Therefore, CRN, VC, SSP coupling, and VM coupling can only be used for the states with known transition probabilities to improve the correlation of two Markov chains in the simulations.

6. Simulation examples

We design the following simulation-based estimation algorithm using the aforementioned coupling methods. In this algorithm, only M potentials (or PRFs) $\widetilde{g}(i^*,j), j = 1,\ldots,M$ should be estimated, whereas other potentials (or PRFs) can be obtained from their relationship.

(1) Initialize the value $\widetilde{g}_0 = \mathbf{0}$, i.e., $\widetilde{g}_0(i,j) = 0, i,j \in S$ and set $n = 0$.

(2) Compute $d_n = \widetilde{f} - \mathcal{L}\widetilde{g}_n = \widetilde{f} - \widetilde{g}_n + \widetilde{P}\widetilde{g}_n$ (when $\widetilde{P}$ is unknown, $\widetilde{P}\widetilde{g}_n(\widetilde{X}_l^k, \widetilde{Y}_l^k)$ may be replaced by $\widetilde{g}_n(\widetilde{X}_{l+1}^k, \widetilde{Y}_{l+1}^k)$ as implemented in Munos (2006), which will introduce an additional variance). Select a recurrent reference state i^* and simulate K sample paths $(\widetilde{X}_l^k, \widetilde{Y}_l^k)$ with $\widetilde{X}_0^k = i^*, \widetilde{Y}_0^k = j$, $l = 0,1,\ldots,L_{i^*j}^k$, $k = 1,2,\ldots,K, j \in S$, by using the coupled transition probability matrix $\widetilde{P}$. Then, perform the following update:

$$\widetilde{g}_{n+1}(i^*,i^*) = 0,$$

$$\widetilde{g}_{n+1}(i^*,j) = \widetilde{g}_n(i^*,j) + \frac{1}{K}\sum_{k=1}^{K}\sum_{l=0}^{L_{i^*j}^k - 1} d_n(\widetilde{X}_l^k, \widetilde{Y}_l^k),$$

$$\widetilde{g}_{n+1}(j,i^*) = -\widetilde{g}_{n+1}(i^*,j),$$

$$\widetilde{g}_{n+1}(i,j) = -\widetilde{g}_{n+1}(j,i) = \widetilde{g}_{n+1}(i^*,j) - \widetilde{g}_{n+1}(i^*,i),$$
$$\forall i \neq i^* \in S,$$

where $L_{i^*j}^k$ denotes the first merging time of the kth sample path.

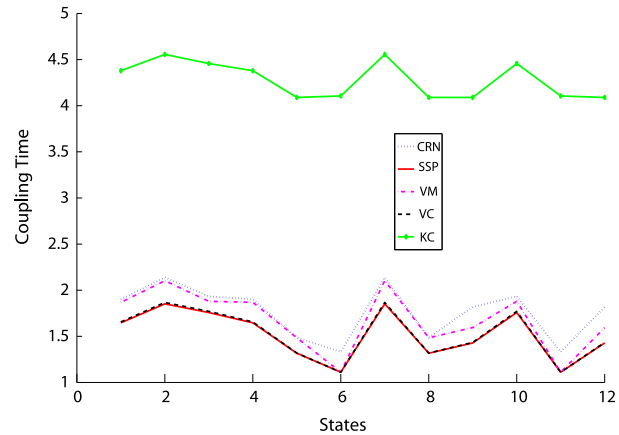

Fig. 1. Coupling time for different coupling methods.

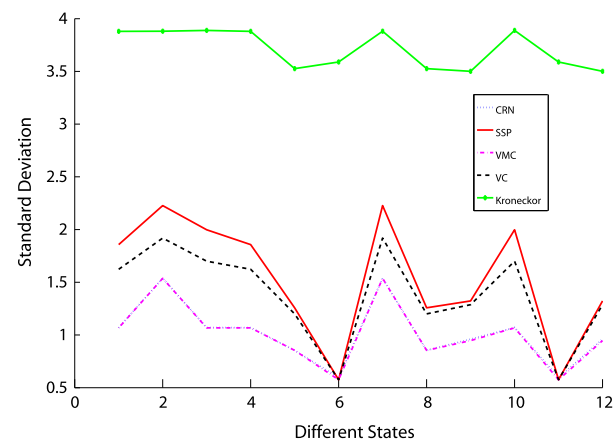

Fig. 2. Variances in different coupling methods.

(3) If $\|\widetilde{g}_{n+1} - \widetilde{g}_n\| \le \varepsilon$, then stop; otherwise, set $n = n+1$, and go to Step 2.

Subsequently, we consider a simple example. Let $\boldsymbol{X}$ be a Markov chain on state space $S = \{1,2,3,4\}$ with the following transition probability matrix:

$$P = \begin{bmatrix} 0.5 & 0.1 & 0.1 & 0.3 \\ 0.2 & 0.3 & 0.3 & 0.2 \\ 0.1 & 0.2 & 0.5 & 0.2 \\ 0.2 & 0.4 & 0.3 & 0.1 \end{bmatrix}.$$

The reward function is $f = (1,2,3,4)^T$.

We compute the coupling (or merging) times and the variance of PRFs for different coupling matrices, which are described in Figs. 1 and 2, respectively. From the results, SSP coupling has the minimal coupling time and the coupling time of VC is extremely close to that of SSP coupling. Moreover, VM coupling has the minimal variance and the variance of CRN coupling is extremely close to that of VM coupling.

We run the estimation algorithm with geometric variance reduction for the aforementioned example. The algorithm is run 20 times for this example by using the method presented in Munos (2006) and different coupled transition probability matrices. In each run, set $K = 100$ in (7) and truncation parameter $L^* = 100$ in (10). Moreover, we assume that average reward and steady state distribution are obtained beforehand in the method presented in Munos (2006) and the PRFs are known in VM coupling. The mean

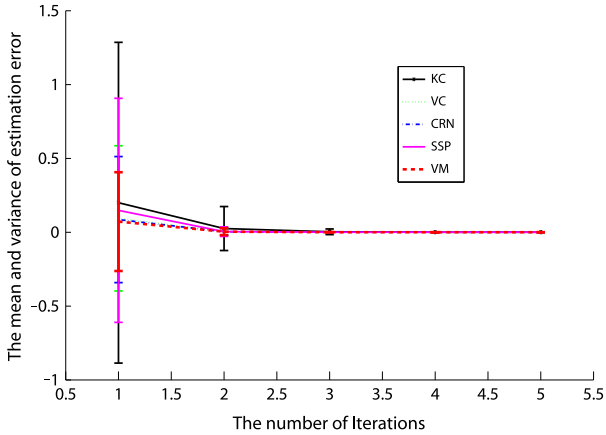

Fig. 3. Mean and variance under different coupling methods.

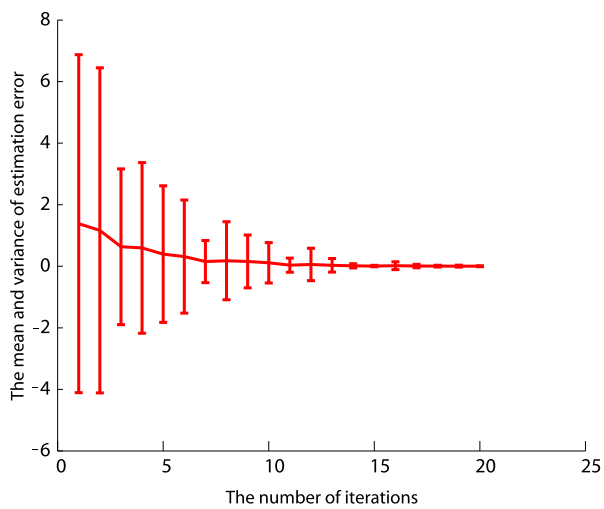

Fig. 4. Simulation result of the method in Munos (2006).

errors between estimates and the true value of PRF $d(1,3)$ under different coupled transition matrices are described in Fig. 3 and the simulation results of the method in Munos (2006) for $g(3)$ is described in Fig. 4. In these figures, a vertical line segment denotes the variance of 20 estimates. The improved algorithms provide a good estimation of PRF after three iterations and the variance is reduced with a geometric rate. However, the method in Munos (2006) requires 18 iterations and the variance reduction rate does not decreases monotonously. Moreover, the method in Munos (2006) produces a larger variance in each iteration. From Fig. 3, VM coupling provides the minimal variance and CRN coupling provides a simple coupling method for variance reduction. From Table 1, SSP coupling may minimize the number of the simulation steps and VC provides a simple coupling method for reducing simulation steps. In each iteration, the simulation steps from each state in the method in Munos (2006) are L^* and K sample paths are simulated. However, the simulation steps from each state in the coupling-based methods are determined by coupling time and K-coupled sample paths are simulated. To guarantee good estimates of π and η, we set $K = 100$ and $L^* = 100$. The simulation steps from each state in five iterations are $5KL^* = 50\,000$ in the method in Munos (2006). In the coupling-based methods, the simulation steps from each state in five iterations are approximately $5KT^*$, where T^* is the average coupling time of the coupled sample path.

Table 1
Average number of simulation steps from each state in five iterations.

KC	CRN	VC	VM	SSP	Method in Munos (2006)
2136	884	762	836	756	$5KL^*$

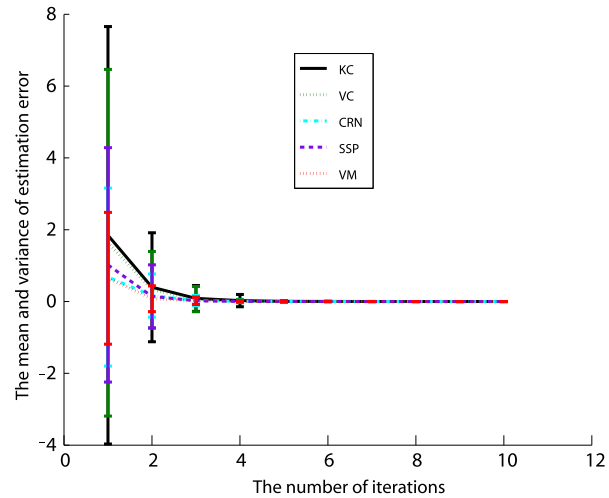

Fig. 5. Mean and variance under different coupling methods.

Finally, we consider a Markov chain with large state space, i.e., a birth and death process with a state space $S = \{1, 2, \ldots, M\}$, and the following transition probability matrix:

$$P = \begin{bmatrix} 1-\theta & \theta & \cdots & 0 & 0 \\ 1-\theta & 0 & \theta & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1-\theta & \theta \end{bmatrix},$$

where $0 < \theta < 1$ is a parameter. Moreover, we assume that the reward function is $f(i) = i$, $i \in S$. For this example,we simulate the algorithm with geometric variance reduction based on different coupling methods and compare the total times of computation and simulation. The algorithm is run 20 times and $K = 200$. The case of variance reduction for PRF $d(1,3)$ when $\theta = 0.5$ and $M = 10$ is shown in Fig. 5, in which VM coupling provides the optimal variance reduction with a geometric rate. From Fig. 6, the sums of the computation and simulation times are different when θ takes $0.1, 0.2, \ldots, 0.9$ and $M = 10$ is fixed. All the coupling methods requires less time than the methods in Munos (2006). When $\theta = 0.5$, an obvious difference is observed among different coupling methods, in which SSP coupling and VM coupling require less time. Fig. 7 presents the sum of the computation and simulation times with respect to state number M when $\theta = 0.5$. From these results, although SSP coupling and VM coupling requires additional computation amount to solve the linear programming problems, they do not increase the total time of the estimation algorithm. In particular, SSP coupling requires the least simulation time among all the coupling methods, and thus, significantly reduce simulation cost.

7. Conclusion

In this study, we present novel approaches for the EGVR of average reward performance potential by combining PRFs and coupling methods. These approaches do not require estimating average reward and truncating performance potential. The coupling methods may further reduce variance or simulation time and effectively guarantee geometric variance reduction can be

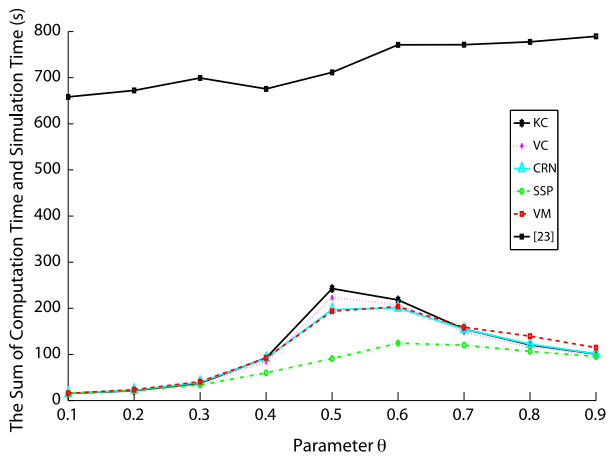

Fig. 6. Comparison of run times for different θ values.

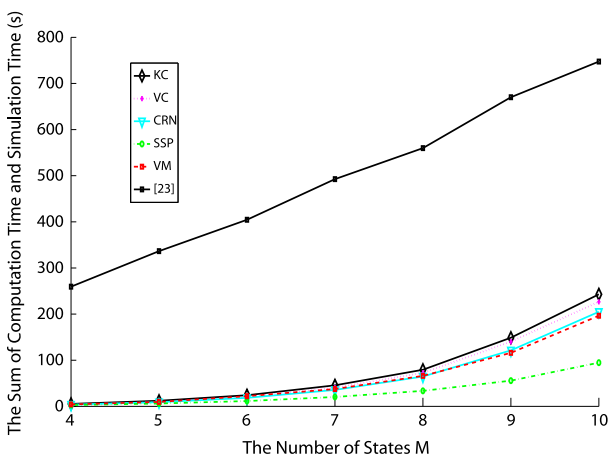

Fig. 7. Comparison of run times for different M values.

guaranteed effectively. In particular, we provide two optimal coupling methods. SSP coupling method provides the optimal coupled transition matrix to minimize the number of simulation steps, whereas VM coupling method provides the optimal coupled transition matrix to minimize PRF variance. Although these two coupling methods are required to solve linear programming problems and increase "thinking" time, they reduce variance and simulation time and improve simulation efficiency. All the coupling methods can also be applied to regular Monte Carlo simulations to estimate PRFs and enhance the efficiency of simulations. However, several questions still need to be considered further . For example, how to select reference state i^* to enhance estimation performance. Moreover, the algorithm complexity analysis and the extension of coupling-based estimation methods to continuous time Markov process (Guo & Hernández-Lerma, 2009), Semi-Markov processes (Huang, Guo, & Song, 2011; Tang, Yin, & Xi, 2007), and hybrid stochastic systems (Kang, Zhai, Liu, Zhao, & Zhao, 2014) are expected.

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